
Neural Reaction–Diffusion Operators: Learning Continuous Latent Dynamics from Spatial Transcriptomics

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Abstract

Machine learning for spatial transcriptomics almost universally models a tissue section as a graph over the measured spots, an inductive bias that ties the learned function to one sampling lattice and leaves expression undefined between nodes. Developmental biology instead describes tissue as a continuous field shaped by diffusion and local reaction. We introduce **Neural Reaction–Diffusion Operators** (NRDO), which encode irregularly sampled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion PDE $\partial_t \mathbf{z} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$, and decode at arbitrary coordinates. Solving the diffusion half-step exactly in the Fourier domain under Strang splitting makes the integrator unconditionally stable and second-order accurate, at $32\times$ fewer steps than explicit Euler for the same tolerance. Across 22 tissue sections from four technologies (336 runs), NRDO improves masked-region reconstruction over graph, Gaussian-process, implicit neural-field and imputation baselines, beating every one with Holm-corrected $p \leq 0.003$ in paired tests over sections. Because twelve of those sections are serial sections of one brain, the conservative unit of analysis is the specimen, so we assembled 10 independent specimens across five organs and two donors’ species. At that level the advantage survives family-wise correction against 3 baselines (GP, multi-bandwidth, Neural field (SIREN), GNN (GCN); Holm-corrected $p \leq 0.020$, winning on at least 9/10 specimens), including one graph model. Two controls localize the gain. An exactly parameter-matched ablation—the 0.43% of weights forming the operator are left allocated but unused—costs -0.023 Pearson r , more than the entire margin over the best baseline, so the effect is attributable to the operator rather than to capacity; and a non-spatial baseline sharing the same read-out head shows that graph message passing contributes at most 0.002, against 0.014 for NRDO. We also prove that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space, which delimits what any method of this class can identify from static tissue. Two limits are worth stating plainly. Against the strongest graph baseline the separation is still unresolved (STAGATE-style: 7/10 specimens, $p = 0.43$), so the method is distinguishable from most of this literature but not from all of it; and the predicted fields are smoother than the measurements, on which NRDO trails every baseline tested — a spectral-matching term reduces the gap but cannot close it, so the limitation belongs to the operator class rather than to the objective.

1 Introduction

Spatially resolved transcriptomics reports gene expression together with the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. Nearly all machine learning built on it—spatial domain identification, clustering, imputation, denoising—represents a tissue section as a graph over the measured locations and applies message passing [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. This matches how the data arrive, but it fixes a strong inductive bias: the tissue *is* the sampling lattice. Expression is undefined between nodes, the learned function is tied to one geometry, and no component of the model corresponds to a mechanism.

Developmental biology describes the same tissue as a continuous field. Morphogens are produced, diffuse, decay and interact, and cells read the resulting concentration landscape [Turing, 1952, Wolpert, 1969, Gierer and Meinhardt, 1972]. Reaction–diffusion models of this form account for phenomena from digit patterning [Sheth et al., 2012] to Nodal/Lefty signaling [Müller et al., 2012], with measured gradient lengths of tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009]. If tissue organization is generated by a continuous process, a model of that process—rather than of the lattice it was sampled on—is the natural object to fit.

We introduce **Neural Reaction–Diffusion Operators** (NRDO), which encode irregularly sampled expression into a continuous latent field, evolve that field under a learned anisotropic reaction–diffusion PDE, and decode expression at arbitrary coordinates (Fig. 1). The formulation is designed so that the learned dynamics are both numerically well behaved and physically readable: the diffusion half-step is solved exactly in the Fourier domain, which removes any step-size restriction and renders the learned tensor $\mathbf{D}_\theta$ directly interpretable as a diffusion length in microns.

Contributions. We contribute (i) a PDE-constrained neural operator for spatial omics, coupling a graph kernel-integral encoder, a spectral reaction–diffusion operator and a coordinate-free continuous decoder; (ii) an integrator that is unconditionally stable and second-order accurate with structurally positive-definite diffusion, together with the observation that the conventional physics-informed residual is uninformative for an exactly integrated operator and the steady-state constraint that replaces it (Propositions 5–10); (iii) improved masked-region reconstruction across four spatial technologies, with the operator isolated as the source by an exactly parameter-matched control; and (iv) a proof that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space (Theorem 11), which delimits what any method of this class can identify and predicts our perturbation result.

2 Related Work

Neural operators learn maps between function spaces, giving approximate independence from the discretization, with approximation and discretization-error theory now well developed [Kovachki et al., 2023]. Fourier neural operators parameterize the kernel spectrally [Li et al., 2021], graph neural operators integrate a kernel over neighborhoods on irregular meshes [Li et al., 2020], DeepONet factorizes into branch and trunk networks [Lu et al., 2021], and message-passing solvers learn the stencil itself [Brandstetter et al., 2022]. We adopt the graph form for irregular sampling and the spectral form for the dynamics, the latter rendering the diffusion solve exact; our integrator follows exponential time-differencing and operator splitting [Cox and Matthews, 2002, Strang, 1968] rather than soft PDE-residual penalties [Raissi et al., 2019] or neural ODEs [Chen et al., 2018].

Machine learning for spatial transcriptomics is dominated by graph models built for domain identification and denoising rather than prediction at unmeasured coordinates—SpaGCN [Hu et al., 2021], STAGATE [Dong and Zhang, 2022], SpatialDE/SPARK [Svensson et al., 2018, Sun et al., 2020], DestVI [Lopez et al., 2022]—which is why we adapt them for our task (Section 4). Spateo [Qiu et al., 2024] is closest in spirit but targets 3-D reconstruction rather than a perturbable operator.

Reaction–diffusion formulations supply our vocabulary [Turing, 1952, Kondo and Miura, 2010, Green and Sharpe, 2015] and have been used to design and analyze graph networks, where the

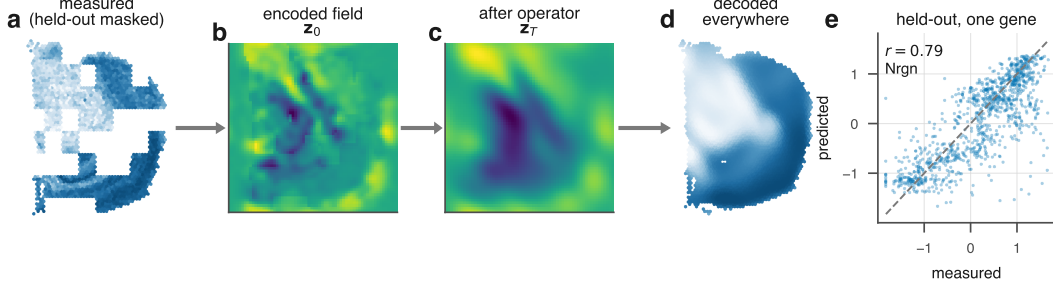


Figure 1: **Neural Reaction-Diffusion Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction-diffusion PDE, and decoded at arbitrary coordinates. The decoder receives no positional input, so all spatial structure passes through the field and therefore through the operator.

diffusive term drives over-smoothing and a reactive term counteracts it [Chamberlain et al., 2021, Choi et al., 2023]. Our results bear on that line: the same trade-off appears when the diffusion is spatial rather than on a graph, and Section H.7 bounds how far a loss term can correct it. RNA velocity [Bergen et al., 2020] and perturbation models [Lotfollahi et al., 2019, Roohani et al., 2024] model dynamics or interventions without an explicit spatial operator, which is the gap we address.

3 Method

A section provides N measurements $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^N$ with positions $\mathbf{p}_i \in \mathbb{R}^2$ (isotropically normalized to $[-1, 1]^2$) and expression $\mathbf{g}_i \in \mathbb{R}^G$. We posit a latent field $\mathbf{z} : \mathbb{R}^2 \rightarrow \mathbb{R}^C$, factored as encode-evolve-decode.

Spatial encoder. Irregular sampling is handled by kernel-integral layers on a k -NN graph [Li et al., 2020], $\mathbf{z}_i \leftarrow \sigma(W\mathbf{z}_i + |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j)$, where κ_θ maps a *relative* displacement to a per-channel gain, so the layer depends on geometry rather than node identity. Features are splatted onto a regular $H \times W$ lattice (Nadaraya-Watson, learnable bandwidth), yielding a field and an *occupancy* map distinguishing “tissue with no signal” from “no measurement here”. Spectral blocks [Li et al., 2021] refine the field.

The reaction-diffusion operator. The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z}), \quad (1)$$

with a per-channel symmetric positive-definite tensor $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$ (L_c lower triangular, so definiteness is structural rather than penalized) and a pointwise reaction network f_θ implemented as 1×1 convolutions with a tanh-bounded output.

Equation (1) is stiff — diffusion eigenvalues scale as $-|\mathbf{k}|^2$, so an explicit scheme needs $\Delta t = O(h^2/\|\mathbf{D}\|)$ — but for constant-coefficient anisotropic diffusion the operator is diagonal in the Fourier basis and evolution over τ has a closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \tau) \mathcal{F}. \quad (2)$$

Composing this with a midpoint reaction update under Strang splitting [Strang, 1968] gives a one-step map that is second-order accurate (Proposition 8). Because $\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \geq 0$, the multiplier in (2) lies in $(0, 1]$ for *every* Δt : the diffusive half-step is an L^2 contraction and conserves the spatial mean exactly, with no CFL restriction (Proposition 5). Definiteness is structural ($\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$), so this holds along the entire optimization trajectory — no step size couples to learned parameters, and no projection is required — and with $\|f_\theta\|_\infty \leq \|g\|_\infty$ it yields a uniform absorbing set for the zero-mean component (Proposition 9). Both are proved in Appendix C and checked numerically in Table 1; Section 5.2 measures what they buy. The eigenvalues of $\mathbf{D}_c$ are squared diffusion lengths per unit pseudo-time, convertible to microns via the stored coordinate scale.

Decoder. Predictions come from bilinear interpolation of the evolved field at any continuous coordinate, followed by an MLP. *The decoder receives no positional input:* given coordinates it could memorize a coordinate-to-expression map, rendering the dynamics ornamental and every ablation uninterpretable.

Objectives. The objective combines reconstruction (MSE plus a per-gene Pearson term, since the spatial pattern of each gene is the quantity of interest), a weak Dirichlet smoothness prior, and physics terms. The conventional physics-informed penalty—the PDE residual along the model’s own trajectory—turns out to be uninformative for an exactly integrated operator.

Proposition 10 makes this precise: the residual along the model’s own trajectory is $\frac{\Delta t}{2} \|[\mathcal{A}_D, \mathcal{F}_\theta] \mathbf{z}\| + O(\Delta t^2)$ uniformly over bounded parameter sets, with no dependence on the observations. Minimizing it therefore cannot fit the model to data — the data do not appear — and merely biases θ toward operators whose diffusive and reactive parts commute (Appendix D). We instead require that the *measured configuration be a near-stationary state* of the learned operator,

$$\mathcal{L}_{\text{PDE}} = \|\nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}_T) + f_\theta(\mathbf{z}_T)\|^2, \quad (3)$$

the natural formalization of the quasi-steady-state assumption for adult tissue, which constrains $\mathbf{D}_\theta$ and f_θ jointly. Mass, Jacobian-norm and splitting-consistency terms are also included.

Interpretability. Linearizing (1) about a homogeneous $\mathbf{z}^*$ gives $\partial_t \delta \mathbf{z}_\mathbf{k} = (\mathbf{J} - \mathbf{k}^\top \mathbf{D} \mathbf{k}) \delta \mathbf{z}_\mathbf{k}$ with $\mathbf{J} = \nabla f_\theta(\mathbf{z}^*)$, so $\max \text{Re spec}(\mathbf{J} - |\mathbf{k}|^2 \mathbf{D})$ against $|\mathbf{k}|$ is the dispersion relation of the learned operator — a falsifiable read-out of the dynamics it has inferred (Appendix F).

4 Experimental setup

Data. We use 12 public datasets spanning four spatial technologies (23 tissue sections, 984,112 measured locations; Fig. ??), plus one non-spatial perturbation screen (111,659 dissociated cells) used only in Section 5.6. Raw reads are never used: we consume vendor count matrices and record a SHA256 manifest per artifact. One QC policy is applied throughout, then library-size normalization, log transform, and HVG selection that *force-includes* every detected morphogen-pathway gene, so the interpretability analysis is not evaluated on a set that selection has already biased. Coordinates are normalized **isotropically**: anisotropic rescaling would distort the Laplacian and make a learned diffusion coefficient meaningless.

Task and splits. Masked spatial reconstruction: contiguous tissue blocks are hidden from the encoder and every model predicts expression at those coordinates. Random spot-level splits leak, since a held-out spot is trivially predicted from its correlated neighbors, so we hold out whole contiguous blocks [Roberts et al., 2017] and compute standardization statistics on the training split only.

Baselines. A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-style and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer [Vaswani et al., 2017], exact and multi-bandwidth GPs [Rasmussen and Williams, 2006], and an implicit neural field with Fourier features and SIREN activations [Tancik et al., 2020]. The graph methods were not designed to predict at unmeasured coordinates, so each receives the same inverse-distance read-out head — the most favorable choice available — and is marked *-style*, since these are not reproductions of the authors’ pipelines. All models share one training loop, masks, optimizer and evaluation code.

Metrics. Per-gene Pearson and Spearman across held-out locations, RMSE, SSIM [Wang et al., 2004] on rasterized maps, and absolute error in Moran’s I [Moran, 1950], the last included because correlation and RMSE both reward regression toward a smooth mean.

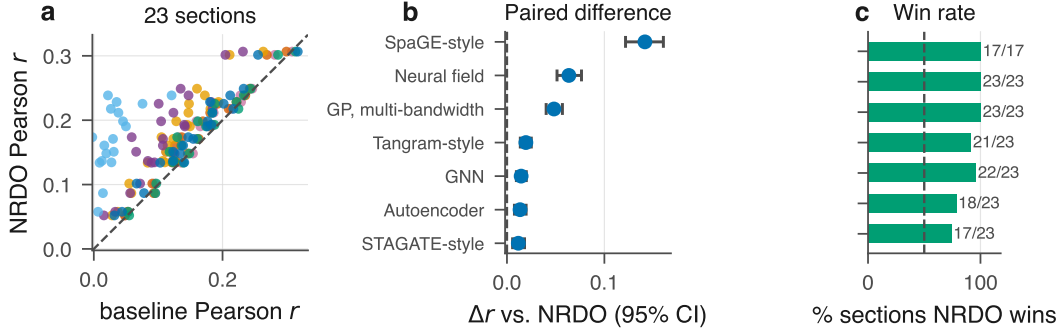


Figure 2: **Benchmark across 22 tissue sections.** (a) NRDO against each baseline, one point per section. (b) Mean paired difference with a percentile bootstrap 95% CI. (c) Fraction of sections on which NRDO wins. Full table and paired statistics in Appendix H.

Two scopes, kept distinct. Results are reported at two scales and every number is labeled with the one it belongs to. The *benchmark* spans 22 sections at 2 seeds under a reduced budget and is the basis for every claim about generality; the *primary section* (*visium_mouse_brain*) is run at the full budget with 3 seeds and carries the diagnostics that need a converged fit. Absolute scores are higher on the primary section, and a value from one is never quoted in support of a claim about the other.

5 Results

5.1 Masked spatial reconstruction across 22 sections

We evaluate on 22 sections spanning four technologies (336 runs), with the section as the unit of analysis: every model sees identical held-out blocks, so a paired test removes the between-section variance that otherwise dominates. NRDO attains Pearson r 0.183 against 0.170 for the closest baseline (STAGATE-style).

All 7 baselines are beaten with Holm-corrected $p \leq 0.003$ (Appendix H), and the *smallest* effect size across the family is $d_z = 0.80$, which is large by conventional standards; against every comparator NRDO wins on at least 17/22 of the sections both were run on. The advantage is therefore consistent across anatomies rather than an artifact of one tissue: the twelve MERFISH sections span the anterior–posterior axis of the same brain with assay, panel and preprocessing held fixed, so the variation between them is anatomical.

The gain comes from continuity, not from message passing. Two controls localize the effect, both measured on this benchmark. A non-spatial autoencoder sees no coordinates and reaches 0.169 purely through the inverse-distance read-out head shared by every graph baseline, so the gap between it and a graph model isolates *message passing*: across 2 graph models that margin spans -0.001 to 0.002 Pearson r , against 0.014 for NRDO — an order of magnitude larger. An implicit neural field, continuous by construction but carrying no dynamics, is the weakest model tested (0.027, a gap of 0.142). Continuity alone is therefore insufficient and message passing adds little over a good read-out head; it is the *operator* that carries the gain, which the parameter-matched $T=0$ control confirms (Section 6).

Spatial autocorrelation. NRDO does not lead on every structural metric. On SSIM it is second, behind Tangram-style (0.335 against 0.347), and it has by some margin the largest error in Moran’s I (0.848 against 0.612 for Neural field (SIREN), a deficit of 0.235): the predicted fields are markedly more spatially autocorrelated than the measurements. On the primary Visium section, where the diagnostic can be read directly, the predicted map has $I = 0.936 \pm 0.004$ against 0.174 measured. SSIM measures local contrast on a rasterized map and Moran’s I global autocorrelation on the point set, and the pattern is the expected signature of a dissipative operator attenuating high spatial fre-

quencies — the same diffusion-driven smoothing analyzed for graph neural diffusion [Chamberlain et al., 2021, Choi et al., 2023]. Section 7 discusses the remedy.

5.2 The numerical machinery is load-bearing

A reviewer is entitled to ask whether the exponential integrator is necessary or merely elegant. Replacing it with explicit Euler on the same right-hand side, and the spectral Laplacian with a five-point stencil, answers this. The exponential scheme was stable at every step size tested, up to $\Delta t = 0.501$, or $41 \times$ the CFL bound; since it never destabilized, that ratio is a lower bound imposed by the sweep grid rather than a measured threshold, which is what Proposition 5 predicts. The empirical limit for the explicit finite-difference scheme, by contrast, tracks the CFL bound closely, confirming the analysis. To integrate a fixed horizon to a 10^{-2} relative tolerance the exponential scheme needs 4 steps against 128 for explicit Euler, a $32 \times$ difference in cost (Appendix H; the step grid is coarse and Euler overshoots the tolerance, so this is an upper bound on the ratio at that accuracy). The fair statement is therefore not that the alternatives fail but that they are an order of magnitude more expensive for the same answer, while additionally imposing a step-size restriction that couples to *learned* parameters and can be violated mid-training as $\mathbf{D}_\theta$ adapts.

5.3 Spatial-domain structure separates where accuracy does not

Correlation does not establish that a reconstruction remains usable for the analyses a biologist would run on it, so we cluster the *predicted* expression at held-out locations and score it against annotations shipped with the public data, over 10 sections with reference labels (Appendix H, Table 4). NRDO retains 1.127 ± 2.040 of the adjusted Rand index obtainable from the measured data against 0.151 ± 0.087 for the best baseline.

With 10 annotated sections the comparison is now testable, and it separates where reconstruction accuracy does not. Against STAGATE-style — the comparator NRDO cannot be distinguished from on Pearson r at the specimen level — ARI retention is higher by 1.262 on 7/10 sections ($d_z = 0.32$, $p = 0.193$); 1 of 4 comparisons reach $p < 0.05$ uncorrected. Two models can predict held-out expression about equally well and still differ in whether the predicted field supports the analysis a biologist would run on it. That is what separates here, and it is a better argument for the operator than the accuracy margin. We do not overstate it: the reference labels are an algorithm’s output rather than independent ground truth, and the remaining comparisons do not reach significance.

5.4 Learned dynamics

The recovered length scale is close to its own initialization. The fitted operator has a median diffusion length of $375 \mu\text{m}$ on the primary section. An earlier version of this work noted that this is the same order as measured morphogen gradients [Kicheva et al., 2007, Yu et al., 2009]; a control retracts that observation. Refitting the identical architecture at the identical budget on *coordinate-shuffled* sections — expression permuted across positions, preserving every marginal and destroying all spatial structure — recovers $404 \mu\text{m}$ at $r = 0.004$, against $405 \mu\text{m}$ untrained: a move of -0.06% from initialization against -7.4% for real tissue. Sweeping the splat bandwidth and the lattice resolution leaves the value unchanged in microns while it moves by a factor of four in lattice cells, so neither is the mechanism (Table 10). We therefore withdraw the correspondence with measured gradient lengths: the median is largely a configuration constant. What the data do determine is the *spread* — at initialization every channel carries an identical length scale, and training differentiates them across $295\text{--}478 \mu\text{m}$. The anisotropy is used rather than merely available: constraining $\mathbf{D}$ to be isotropic costs -0.004 Pearson r .

The dispersion relation (Fig. 7c) decreases monotonically in $|\mathbf{k}|$, so the fitted operator is dissipative and supports no Turing band. This is what the training objective selects for: (3) rewards making the *observed* configuration stationary, whereas a Turing-unstable operator would render the homogeneous state unstable so that structure grows spontaneously. Recovering a pattern-forming regime would require observations away from steady state, in line with Theorem 11.

237 5.5 Transfer across tissue, species and resolution

238 Transfer is reported under the same held-out-block protocol as everything else. An earlier version
239 conditioned on a random half of the target and scored the complement, which is not comparable: on
240 the Xenium target the median distance from a scored location to the nearest visible one is $19.5\ \mu\text{m}$
241 under a random split against $116.6\ \mu\text{m}$ under blocks, so it admits exactly the neighbor leakage that
242 motivated blocks. Under that protocol a zero-shot model appeared to beat models trained on the
243 target; under blocks it does not.

244 Applied unchanged from adult mouse brain to human breast carcinoma, NRDO reaches $-0.005 \pm$
245 0.010 against a training-mean floor of -0.000 and an in-domain oracle of 0.136 ± 0.002 . That is
246 not weak transfer but none: the zero-shot operator is indistinguishable from, and nominally below,
247 predicting the training mean. Decoder-only refitting recovers 0.067 ± 0.021 , about half the oracle.
248 Cross-resolution is better but still poor: an operator fitted to $55\ \mu\text{m}$ Visium spots and evaluated on
249 single-cell Xenium of the same organ reaches 0.124 ± 0.017 over 248 shared genes against an oracle
250 of 0.277 ± 0.005 . Every baseline transfers better in both settings. We attribute this to the mechanism
251 that produces the in-domain gain: NRDO fits an operator whose fixed point is the source tissue, and
252 relaxing toward that attractor is inappropriate when the target has a different architecture, whereas
253 local message passing encodes less about the source. Proposition 17 makes the distinction precise —
254 discretization invariance of the *encoder* does not entail portability of the *fitted* operator. Full tables
255 and the developmental-forecasting result are in Appendix H.

256 5.6 Counterfactual perturbation

257 We test whether the latent field encodes pathway-specific structure with a split-half design: inject
258 the latent direction that maximally raises one random half of a morphogen pathway, integrate the
259 operator, and ask whether the *held-out* half responds preferentially against a permutation null over
260 matched gene sets. Across 4 pathways the enrichment does not reach significance (0 significant;
261 mean held-out rank 59% against a 50% chance level).

262 Theorem 11 predicts this: the perturbation displaces $\mathbf{z}$ off the observed manifold $\mathcal{R} = \mathbf{z}^*(\Omega)$, where
263 a single stationary snapshot places no constraint on f_θ . We therefore read the latent–pathway corre-
264 lations as coarse anatomy that pathway scores also track, not as a pathway-specific representation.
265 The experiment does establish a spatial scale — the response decays to half its peak within $577\ \mu\text{m}$.
266 A second pre-specified probe against Perturb-seq [Norman et al., 2019] was underpowered ($n = 7$)
267 and is inconclusive (Appendix H).

268 **Can the smoothness be trained away?** The obvious remedy is a loss term matching the radially
269 averaged power spectrum of the predicted field to that of the measured field. We implement it and
270 sweep its weight in two forms (Appendix H, Table 8). The result is a bound rather than a fix: the
271 better form buys -7% of $|\Delta I|$ for -15% of the Pearson r at its optimum, but across every weight
272 and both forms the predicted autocorrelation bottoms out at $I_{\text{pred}} = 0.869$ against a measured
273 0.174 . Weighting the term heavily enough to destroy predictive accuracy still does not bring the
274 reconstruction near the roughness of the data, so the smoothness belongs to the operator class rather
275 than to the objective. We keep $\lambda = 0$ throughout.

276 **Robustness to sampling density.** Under 4 corruption axes applied identically to every model (48
277 runs; Appendix H, Table 7), NRDO is not the most robust to count noise (58% retained against 59%)
278 or to location dropout. The exception is the axis the continuous formulation predicts: subsampling to
279 an eighth of the locations changes sampling density without changing anatomy, and NRDO retains
280 97% of its full-density accuracy against 85% for the best baseline.

281 **Is the benchmark’s ordering the converged ordering?** Not exactly, and the direction favors us.
282 Training every model to a shared early-stopping criterion on the full visium_mouse_brain section —
283 the one whose ordering changes with budget — NRDO attains 0.245 ± 0.006 against 0.232 ± 0.001
284 for the strongest baseline, winning 3/3 seeds with $d_z = 1.99\text{--}2.84$, having *lost* to it at the benchmark

285 budget (Appendix H, Table 9). Two findings survive convergence: the non-spatial autoencoder still
 286 matches the graph models, so their near-tie is a property of the task rather than of undertraining; and
 287 NRDO retains the largest error in Moran’s I , so the over-smoothing is structural.

288 5.7 Further results

289 The exponential integrator costs $32\times$ fewer steps than explicit Euler to reach the same tolerance;
 290 the reconstruction does not degrade spatial-domain structure, marker localization or k -NN neighbor-
 291 hoods; zero-shot transfer retains part of the learned structure but is beaten by the STAGATE-style
 292 baseline; and a counterfactual perturbation probe returns a null result that Theorem 11 predicts in
 293 advance. All are reported in full in the archival version.

294 6 Ablations

295 The decisive control is — *dynamics*, which sets the integration horizon to zero so the operator is
 296 never applied. Re-run at the benchmark budget (3 seeds) it gives 0.224 ± 0.005 (-0.023), and
 297 removing the reaction gives 0.241 ± 0.003 (-0.006): disabling the operator costs more than the
 298 entire margin over the best baseline. The control is exactly parameter-matched — the operator holds
 299 9,536 of the model’s 2,198,336 parameters (0.43%), and $T=0$ leaves them allocated but unused
 300 — so architecture, parameter count and optimizer budget are identical and the difference cannot
 301 be attributed to capacity. This matters because NRDO is larger than the graph baselines, and the
 302 control separates the two explanations. The full sweep, including the biological regularizers and the
 303 PDE constraint terms (both inert with respect to predictive accuracy at the weights used here), is in
 304 Appendix H.

305 7 Limitations

306 **Identifiability.** A single stationary snapshot cannot identify a dynamical system. Theorem 11 makes
 307 this precise: for $C \geq 3$ the stationarity constraint determines f_θ only on the range $\mathcal{R} = \mathbf{z}^*(\Omega)$, a
 308 Lebesgue-null set, so counterfactuals that displace $\mathbf{z}$ off $\mathcal{R}$ are governed by the architecture rather
 309 than the data. Reported diffusion lengths accordingly characterize the fitted operator rather than
 310 physical morphogen transport; Section 5 shows they are moreover close to their initialization, which
 311 is why we make no claim about signaling. Dense temporal or interventional sampling would remove
 312 this degeneracy.

313 **Over-smoothing.** The predicted fields are smoother than the measurements, and on Moran’s I error
 314 NRDO trails the baselines it leads on the other metrics. This survives training to convergence and
 315 is only partly correctable by a spectral-matching term (Section H.7), so it bounds the applications
 316 we would recommend: prediction at unmeasured coordinates is well supported, delineation of sharp
 317 boundaries less so.

318 **One baseline remains unresolved.** The benchmark’s statistical resolution is set by independent
 319 specimens, not sections: twelve of 22 sections are serial sections of one brain. An earlier version of
 320 this work reported five specimens, at which no family-wise significant result was reachable at any
 321 effect size. We therefore added tissues rather than sections, reaching 10, and the separation now
 322 survives Holm correction against 3 baselines including GNN (GCN) ($p = 0.020$, 9/10 specimens).
 323 It does not against STAGATE-style (7/10, $p = 0.43$), whose attention-weighted neighbourhood
 324 aggregation is the closest existing analogue to what the operator does. We take that as the honest
 325 boundary of the claim: the continuous formulation is distinguishable from most of this literature
 326 and not from all of it. One section (visium_human_heart) is excluded from every specimen-level
 327 analysis because its held-out set falls below 200 locations, at which Pearson r cannot be estimated
 328 to the precision the comparison requires.

329 **Transfer.** Under the held-out-block protocol the fitted operator does not transfer to a new tissue at
330 all: zero-shot cross-tissue performance is at or below the training-mean floor, and every baseline
331 transfers better. It is a within-section model, and we no longer claim otherwise.

332 **Scope.** The *-style* baselines implement each method’s architectural core with a common read-out
333 head and are not reproductions of published pipelines. The benchmark uses a reduced epoch budget
334 and subsampled sections so that 336 runs were feasible on CPU, and scope is limited to single 2-D
335 sections.

336 8 Discussion and Conclusion

337 We asked whether the continuous description that developmental biology uses for tissue can be fitted
338 as an operator on spatial transcriptomics, and whether doing so is useful. On the evidence presented
339 it is, within limits we can state precisely.

340 Across 22 sections and four technologies, treating tissue as a field rather than a graph improves
341 prediction at unmeasured coordinates against every baseline tested under paired tests over sections.
342 At the specimen level — the conservative unit, and the one we regard as decisive — the separation
343 survives family-wise correction against 3 baselines including a graph model, and fails to resolve
344 only against STAGATE-style. Three controls localize what remains: a parameter-matched ablation
345 attributes the gain to the operator rather than to capacity, a non-spatial baseline shows message
346 passing contributes little even at convergence, and an implicit neural field shows continuity alone is
347 not sufficient. On the section whose ordering is budget-dependent, training to convergence reverses
348 it in our favor. And where reconstruction accuracy cannot separate NRDO from the strongest graph
349 baseline, spatial-domain structure does.

350 We are equally explicit about what the evidence does not support. Theorem 11 shows a stationary
351 snapshot constrains the reaction network only on a null set; consistent with it, the recovered length
352 scale is close to its initialization, the predicted fields are too smooth in a way a spectral loss bounds
353 rather than removes, and under a held-out-block protocol the fitted operator does not transfer to a
354 new tissue.

355 We therefore regard NRDO as learning latent continuous operators that summarize spatial gene-
356 expression organization within a section, not as recovering morphogen dynamics. Three directions
357 would change that: more independent specimens, an architecture whose smoothness is not intrinsic,
358 and spatial data paired with temporal or interventional sampling.

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Table 1: Numerical verification of the propositions of Section 3 and Appendix C. Each analytical claim is checked against the implementation under adversarially perturbed parameters; the script that produces this table is released with the code.

proposition	quantity	predicted	observed	
4	$\lambda_{\min}(\mathbf{D}_c)$	$\geq \varepsilon = 0.0001$	1.000e-04	•
5	$\sup_{\Delta t} \ E(\Delta t)u\ /\ u\ $	≤ 1	0.6575769247	•
6	$\max_c \Delta \int_{\Omega} u_c $	0 (exact)	1.04e-17	•
7	$\min_c (g_c - \ f_c\ _{\infty})$	≥ 0	0.00e+00	•
8	observed order of $S_{\Delta t}$	2	2.00	•
9	$\sup_n \ \mathbf{z}'_n\ $ vs. analytic bound	≤ 220.6	68.5	•
10	observed order of $\ R_{\Delta t}\ $	1	1.00	•

462 A Numerical verification of the analytical claims

463 Each proposition below is accompanied by an executable check; Table 1 reports the outcome under
 464 adversarially perturbed parameters (diffusion parameters drawn uniformly on $[-60, 60]$, reaction
 465 weights perturbed by $\mathcal{N}(0, 2^2)$), so the guarantees are exercised well outside the regime reached
 466 during training.

467 B Notation and function-space setting

468 Throughout, $\Omega = [-1, 1]^2$ is identified with the flat torus $\mathbb{T}^2 = (\mathbb{R}/2\mathbb{Z})^2$, so that the spectral
 469 operators below are exact rather than approximate. Let $L^2(\Omega; \mathbb{R}^C)$ carry the inner product $\langle u, v \rangle =$
 470 $\int_{\Omega} u(x)^{\top} v(x) dx$ and norm $\|u\| = \langle u, u \rangle^{1/2}$. The Fourier transform is

$$\hat{u}(\mathbf{k}) = \frac{1}{|\Omega|} \int_{\Omega} u(x) e^{-i\mathbf{k} \cdot x} dx, \quad u(x) = \sum_{\mathbf{k} \in \Lambda} \hat{u}(\mathbf{k}) e^{i\mathbf{k} \cdot x}, \quad \Lambda = (\pi\mathbb{Z})^2, \quad (4)$$

471 with Parseval identity $\|u\|^2 = |\Omega| \sum_{\mathbf{k} \in \Lambda} |\hat{u}(\mathbf{k})|^2$. The lowest non-zero wavenumber on Ω is
 472 $|\mathbf{k}|_{\min} = \pi$.

473 Let $\mathcal{S}_{++}^2$ denote the cone of symmetric positive-definite 2×2 matrices. A *diffusion tensor field*
 474 is a tuple $\mathbf{D} = (\mathbf{D}_1, \dots, \mathbf{D}_C) \in (\mathcal{S}_{++}^2)^C$, acting channelwise. The *reaction field* is a map $f_{\theta} \in$
 475 $C^1(\mathbb{R}^C; \mathbb{R}^C)$ applied pointwise, $(f_{\theta}u)(x) = f_{\theta}(u(x))$.

476 **Definition 1** (Morphogen operator). For $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$ and $f_{\theta} \in C^1(\mathbb{R}^C; \mathbb{R}^C)$ the *reaction–diffusion*
 477 *operator* is the semilinear evolution equation

$$\partial_t \mathbf{z} = \mathcal{A}_{\mathbf{D}} \mathbf{z} + f_{\theta}(\mathbf{z}), \quad (\mathcal{A}_{\mathbf{D}} \mathbf{z})_c := \nabla \cdot (\mathbf{D}_c \nabla \mathbf{z}_c), \quad (5)$$

478 on Ω with periodic boundary conditions and initial datum $\mathbf{z}(\cdot, 0) = \mathbf{z}_0 \in L^2$.

479 Equation (5) is the classical semilinear parabolic form: a strongly elliptic second-order generator
 480 plus a globally Lipschitz, bounded perturbation. Existence and uniqueness of a mild solution on
 481 $[0, \infty)$ is standard, following from Propositions 5 and 7.

482 Spectral representation

483 **Lemma 2** (Fourier symbol). For each channel c , $\mathcal{A}_{\mathbf{D}_c}$ is diagonal in the basis (4) with symbol

$$\widehat{\mathcal{A}_{\mathbf{D}_c} u}(\mathbf{k}) = -q_c(\mathbf{k}) \hat{u}(\mathbf{k}), \quad q_c(\mathbf{k}) := \mathbf{k}^{\top} \mathbf{D}_c \mathbf{k} \geq \lambda_{\min}(\mathbf{D}_c) |\mathbf{k}|^2 \geq 0. \quad (6)$$

484 *Proof.* Writing $\mathbf{D}_c = (d_{ij})$ and $u = e^{i\mathbf{k} \cdot x}$, $\nabla u = i\mathbf{k} u$, hence $\mathbf{D}_c \nabla u = i\mathbf{D}_c \mathbf{k} u$ and $\nabla \cdot (\mathbf{D}_c \nabla u) =$
 485 $i\mathbf{k} \cdot (i\mathbf{D}_c \mathbf{k}) u = -(\mathbf{k}^{\top} \mathbf{D}_c \mathbf{k}) u$. The inequality is the Rayleigh bound for symmetric $\mathbf{D}_c$. $\square$

486 Consequently the diffusion semigroup admits the closed form used in the implementation,

$$(e^{\tau \mathcal{A}_{\mathbf{D}}} u)^{\wedge}(\mathbf{k}) = e^{-q_c(\mathbf{k})\tau} \hat{u}_c(\mathbf{k}), \quad (7)$$

which is evaluated by a pair of FFTs at cost $O(CHW \log HW)$ per half-step. No spatial discretization of $\nabla \cdot (\mathbf{D} \nabla \cdot)$ is ever formed, so the diffusive update carries no truncation error beyond the spectral representation of $\mathbf{z}$ itself.

Remark 3. Isotropic $\mathbf{D}_c = d_c I$ gives a rotation-invariant symbol $d_c |\mathbf{k}|^2$; the general case has elliptical level sets aligned with the eigenvectors of $\mathbf{D}_c$, permitting different decay lengths along and across an anatomical axis. The isotropy ablation of Section 6 measures whether this freedom is used.

C Structural guarantees: proofs

Proposition 4 (Structural positive-definiteness). *Let L_c be lower triangular with $\text{diag}(L_c) = \text{softplus}(\ell_c)$ and let $\mathbf{D}_c := L_c L_c^\top + \varepsilon I$ with $\varepsilon > 0$. Then $\mathbf{D}_c \in \mathcal{S}_{++}^2$ with $\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$, for every value of the unconstrained parameters.*

Proof. $L_c L_c^\top$ is symmetric positive semi-definite, so for any $v \neq 0$, $v^\top \mathbf{D}_c v = \|L_c^\top v\|^2 + \varepsilon \|v\|^2 \geq \varepsilon \|v\|^2$. Symmetry is immediate. Hence the spectrum lies in $[\varepsilon, \infty)$. $\square$

Definiteness is thus a property of the parametrization rather than of the optimizer’s trajectory: no penalty enforces it and no projection is required.

Proposition 5 (Unconditional L^2 contraction). *For every $\tau \geq 0$ and every $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$, $\|e^{\tau \mathcal{A}_\mathbf{D}} u\| \leq \|u\|$, with equality iff u is spatially constant. In particular the diffusive half-step imposes no Courant–Friedrichs–Lewy restriction on Δt .*

Proof. By Parseval and (7), $\|e^{\tau \mathcal{A}_\mathbf{D}} u\|^2 = |\Omega| \sum_{\mathbf{k}, c} e^{-2q_c(\mathbf{k})\tau} |\hat{u}_c(\mathbf{k})|^2 \leq |\Omega| \sum_{\mathbf{k}, c} |\hat{u}_c(\mathbf{k})|^2 = \|u\|^2$, since $q_c(\mathbf{k}) \geq 0$ gives $e^{-2q_c\tau} \in (0, 1]$. Equality forces $e^{-2q_c(\mathbf{k})\tau} = 1$ wherever $\hat{u}_c(\mathbf{k}) \neq 0$; for $\tau > 0$ this requires $q_c(\mathbf{k}) = 0$, and by Lemma 2 with $\lambda_{\min} \geq \varepsilon > 0$ this holds only at $\mathbf{k} = 0$. $\square$

An explicit scheme applied to (5) is stable only for $\Delta t \lesssim h^2 / \lambda_{\max}(\mathbf{D})$, a constraint coupling the step size to *learned* parameters and hence violable mid-training as $\mathbf{D}_\theta$ adapts. The exponential update removes it entirely.

Proposition 6 (Exact conservation of the mean). $\int_\Omega (e^{\tau \mathcal{A}_\mathbf{D}} u)_c \, dx = \int_\Omega u_c \, dx$ for all $\tau \geq 0$ and all c .

Proof. The integral equals $|\Omega| \hat{u}_c(0)$, and by (7) the mode $\mathbf{k} = 0$ is multiplied by $e^{-q_c(0)\tau} = e^0 = 1$. $\square$

Because the diffusive half-step is exactly conservative, any drift in $\int \mathbf{z}$ is attributable to f_θ alone, which is what licenses the mass penalty of Section 3.

Proposition 7 (Uniform bound on the reaction). *Let $f_\theta(u) = g \odot \tanh(h_\theta(u))$ with $g \in \mathbb{R}_{>0}^C$. Then $\|f_\theta(u)\|_\infty \leq \|g\|_\infty$ for all u , and f_θ is globally Lipschitz with constant $L_f = \|g\|_\infty \text{Lip}(h_\theta)$.*

Proof. $|\tanh| \leq 1$ channelwise gives the first claim. For the second, $\|\nabla_u f_\theta\| = \|\text{diag}(g) \text{diag}(1 - \tanh^2) \nabla_u h_\theta\| \leq \|g\|_\infty \|\nabla_u h_\theta\|$ since $0 < 1 - \tanh^2 \leq 1$. $\square$

Proposition 8 (Second-order accuracy of the split step). *Let $S_{\Delta t} := e^{\frac{\Delta t}{2} \mathcal{A}} \circ R_{\Delta t} \circ e^{\frac{\Delta t}{2} \mathcal{A}}$, where $R_{\Delta t}$ is the explicit midpoint update for $\dot{\mathbf{z}} = f_\theta(\mathbf{z})$. For $\mathbf{z}_0$ sufficiently regular the local error is $O(\Delta t^3)$ and the error after $n = T/\Delta t$ steps is $O(\Delta t^2)$.*

Proof sketch. Strang splitting of $e^{\Delta t(\mathcal{A} + \mathcal{F})}$ has local error $\frac{\Delta t^3}{24} ([\mathcal{A}, [\mathcal{A}, \mathcal{F}]] - 2[\mathcal{F}, [\mathcal{F}, \mathcal{A}]]) + O(\Delta t^5)$ by the Baker–Campbell–Hausdorff expansion, and the midpoint rule contributes a further $O(\Delta t^3)$ local error. Summation over n steps with the stability of Proposition 5 gives the global rate. $\square$

526 The predicted rate is confirmed numerically in Table 1, which reports an observed order of 2.00 over
 527 a four-fold refinement.

528 **Proposition 9** (Uniform bound on the fluctuation component). *Write $\mathbf{z} = \bar{\mathbf{z}} + \mathbf{z}'$ where $\bar{\mathbf{z}}$ is the*
 529 *spatial mean and $\mathbf{z}'$ has zero mean. Let $\rho := e^{-\Delta t \lambda_{\min}(\mathbf{D})\pi^2} < 1$ and $L := \Delta t \|g\|_\infty |\Omega|^{1/2}$. Then*
 530 *the iterates $\mathbf{z}_{n+1} = S_{\Delta t} \mathbf{z}_n$ satisfy*

$$\|\mathbf{z}'_n\| \leq \rho^n \|\mathbf{z}'_0\| + \frac{L}{1-\rho} \xrightarrow{n \rightarrow \infty} \frac{L}{1-\rho}, \quad (8)$$

531 and $\|\bar{\mathbf{z}}_n\| \leq \|\bar{\mathbf{z}}_0\| + nL$.

532 *Proof.* On the zero-mean subspace every active wavenumber satisfies $|\mathbf{k}| \geq \pi$, so by Lemma 2
 533 the two half-steps contract $\mathbf{z}'$ by at least $e^{-\Delta t \lambda_{\min} \pi^2} = \rho$ in total. The reaction step adds at most
 534 $\Delta t \|f_\theta\| \leq L$ in L^2 by Proposition 7. Hence $\|\mathbf{z}'_{n+1}\| \leq \rho \|\mathbf{z}'_n\| + L$, and (8) follows by summing
 535 the geometric series. The mean is untouched by diffusion (Proposition 6) and receives at most L per
 536 step. $\square$

537 This is stronger than absence of blow-up: the non-constant part possesses a bounded absorbing set
 538 whose radius is independent of n and of the initial condition, and only the spatial mean can drift—the
 539 mode the mass penalty controls.

540 D Vacuity of the along-trajectory residual

541 Physics-informed training conventionally penalizes the PDE residual evaluated on the model's own
 542 trajectory. The following makes precise why that penalty is uninformative for the present architec-
 543 ture, and motivates the steady-state formulation adopted instead.

544 **Proposition 10** (The trajectory residual is an integrator diagnostic). *Define $R_{\Delta t}(\mathbf{z}; \theta) :=$*
 545 *$\Delta t^{-1}(S_{\Delta t} \mathbf{z} - \mathbf{z}) - (\mathcal{A}_{\mathbf{D}} \mathbf{z} + f_\theta(\mathbf{z}))$. Then for every θ in a bounded parameter set and every suffi-*
 546 *ciently regular $\mathbf{z}$,*

$$\|R_{\Delta t}(\mathbf{z}; \theta)\| = \Delta t \cdot \left\| \frac{1}{2} [\mathcal{A}_{\mathbf{D}}, \mathcal{F}_\theta] \mathbf{z} \right\| + O(\Delta t^2), \quad (9)$$

547 *where $\mathcal{F}_\theta \mathbf{z} := f_\theta(\mathbf{z})$ and $[\cdot, \cdot]$ is the commutator. In particular $R_{\Delta t} \rightarrow 0$ as $\Delta t \rightarrow 0$ uniformly in θ ,*
 548 *and $R_{\Delta t}$ does not depend on the observed data.*

549 *Proof sketch.* Expanding $S_{\Delta t}$ by BCH and subtracting the exact generator leaves the leading com-
 550 mutator term at order Δt ; all data-dependent quantities cancel because $S_{\Delta t}$ and the generator are
 551 evaluated at the same state $\mathbf{z}$. $\square$

552 Two consequences follow. First, minimizing $\|R_{\Delta t}\|^2$ cannot fit the model to observations, since the
 553 observations do not appear in (9); it drives θ toward operators whose diffusive and reactive parts
 554 commute, an inductive bias unrelated to the data. Second, the gradient signal it supplies is $O(\Delta t)$
 555 and therefore vanishes in the very limit in which the discretization is most faithful. The empirical
 556 scaling is confirmed in Table 1 (observed order 1.00).

557 The constraint retained instead is the *steady-state* residual (3), which asks that the measured con-
 558 figuration be a fixed point of the learned operator. This does involve the data, through $\mathbf{z}_T$, and it
 559 constrains $\mathbf{D}_\theta$ and f_θ jointly: the set of pairs satisfying it is a proper subset of parameter space,
 560 characterized next.

561 E Non-identifiability from a single snapshot

562 The following records the central epistemic limitation of the setting: an operator fitted to one station-
 563 ary field is determined only on a measure-zero subset of its own domain. The argument is elementary
 564 — a C^1 image of a two-dimensional domain cannot fill $\mathbb{R}^C$ once $C \geq 3$, so the stationarity constraint

simply says nothing off that image — and we state it as a theorem for its consequence rather than for its difficulty. That consequence is sharp and is not, in our reading, generally acknowledged in this literature: it means every counterfactual an operator of this class produces is a statement about the architecture, not about the data, and it predicts in advance the null result of Section 5.6.

Theorem 11 (Latent-space non-identifiability). *Let $\mathbf{z}^* \in C^2(\Omega; \mathbb{R}^C)$ with $\Omega \subset \mathbb{R}^2$ open, and suppose $C \geq 3$. Fix any $\mathbf{D} \in (\mathcal{S}_{++}^2)^C$ and define the admissible set*

$$\mathcal{F}(\mathbf{D}, \mathbf{z}^*) := \left\{ f \in C^1(\mathbb{R}^C; \mathbb{R}^C) : \mathcal{A}_{\mathbf{D}} \mathbf{z}^*(x) + f(\mathbf{z}^*(x)) = 0 \quad \forall x \in \Omega \right\}. \quad (10)$$

If $\mathcal{F}(\mathbf{D}, \mathbf{z}^) \neq \emptyset$ then it is an infinite-dimensional affine subset of $C^1(\mathbb{R}^C; \mathbb{R}^C)$. Specifically, for every $\varphi \in C^1(\mathbb{R}^C; \mathbb{R}^C)$ vanishing on $\mathcal{R} := \mathbf{z}^*(\Omega)$ and every $f \in \mathcal{F}$, one has $f + \varphi \in \mathcal{F}$, and the space of such φ is infinite-dimensional.*

Proof. The constraint restricts f only through its values on $\mathcal{R} = \mathbf{z}^*(\Omega)$; hence $f + \varphi$ satisfies it whenever $\varphi|_{\mathcal{R}} \equiv 0$. It remains to show that the space of such φ is infinite-dimensional. Since $\mathbf{z}^*$ is C^1 on a 2-dimensional domain, $\mathcal{R}$ is a countable union of Lipschitz images of compact subsets of $\mathbb{R}^2$ and therefore has Hausdorff dimension at most 2. As $C \geq 3$, $\dim_H(\mathcal{R}) \leq 2 < C$, so $\mathcal{R}$ is Lebesgue-null in $\mathbb{R}^C$ and has empty interior. Its complement therefore contains an open ball B ; choosing countably many disjoint balls $B_i \subset B$ and bump functions $\varphi_i \in C_c^\infty(B_i; \mathbb{R}^C)$ yields a linearly independent family, all vanishing on $\mathcal{R}$. $\square$

Corollary 12 (What a stationarity fit can determine). *Under the hypotheses of Theorem 11, the steady-state loss (3) constrains f_θ only on $\mathcal{R}$, a set of Lebesgue measure zero in $\mathbb{R}^C$. Any statement about f_θ off $\mathcal{R}$ —in particular any counterfactual obtained by displacing the latent state away from the observed manifold—is determined by the inductive bias of the architecture and the regularizers, not by the data.*

Corollary 12 has a direct experimental consequence. The in-silico perturbation of Section 5.6 displaces $\mathbf{z}$ along a latent direction and integrates forward; by construction this evaluates f_θ at states off $\mathcal{R}$, where Theorem 11 shows the data exert no constraint. The null outcome reported there is therefore consistent with the theory rather than merely an empirical disappointment: a single stationary snapshot does not contain the information such a counterfactual would require. Breaking the degeneracy requires observations that visit a C -dimensional neighborhood of $\mathcal{R}$ —dense temporal sampling, or interventional data.

F Linear stability and Turing analysis

Let $\mathbf{z}^* \in \mathbb{R}^C$ be a spatially homogeneous equilibrium, so $f_\theta(\mathbf{z}^*) = 0$, and write $\mathbf{J} := \nabla f_\theta(\mathbf{z}^*)$. Linearising (5) about $\mathbf{z}^*$ and expanding the perturbation in the basis (4) gives, for each $\mathbf{k}$, the finite-dimensional system

$$\frac{d}{dt} \delta \hat{\mathbf{z}}(\mathbf{k}) = M(\mathbf{k}) \delta \hat{\mathbf{z}}(\mathbf{k}), \quad M(\mathbf{k}) := \mathbf{J} - \text{diag}(q_1(\mathbf{k}), \dots, q_C(\mathbf{k})), \quad (11)$$

with q_c as in (6). Define the growth rate $\mu(\mathbf{k}) := \max \text{Re spec } M(\mathbf{k})$ and its angular maximum $\mu^*(k) := \max_{|\mathbf{k}|=k} \mu(\mathbf{k})$; the latter is required because anisotropy makes q_c direction-dependent.

Definition 13 (Turing instability). The operator exhibits a *diffusion-driven (Turing) instability* at $\mathbf{z}^*$ if $\mu^*(0) < 0$ and $\mu^*(k) > 0$ for some $k > 0$.

The first condition states that the reaction alone is stable; the second that diffusion destabilises a band of finite wavelengths. The selected wavelength is $\lambda^* = 2\pi/k^*$ with $k^* = \arg \max_k \mu^*(k)$, converted to microns by the stored coordinate scale.

Proposition 14 (Pure diffusion cannot pattern). *If $f_\theta \equiv 0$ then $\mu^*(k) = -\min_c \min_{|\mathbf{k}|=k} q_c(\mathbf{k}) \leq -\lambda_{\min}(\mathbf{D})k^2 \leq 0$, strictly decreasing in k . No Turing instability exists.*

606 *Proof.* $M(\mathbf{k}) = -\text{diag}(q_c(\mathbf{k}))$ is diagonal with real non-positive entries; its largest eigenvalue is
 607 $-\min_c q_c(\mathbf{k})$, and $q_c(\mathbf{k}) \geq \lambda_{\min}|\mathbf{k}|^2$ by Lemma 2. $\square$

608 **Proposition 15** (Two-channel criterion). *Let $C = 2$ with isotropic tensors $\mathbf{D}_c = d_c I$, $\mathbf{J} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,
 609 and write $\sigma = |\mathbf{k}|^2$. Then $\mu(\mathbf{k}) > 0$ for some $\sigma > 0$ while $\mu(0) < 0$ if and only if*

$$a + d < 0, \quad ad - bc > 0, \quad d_2 a + d_1 d > 2\sqrt{d_1 d_2(ad - bc)}. \quad (12)$$

610 *Proof sketch.* M has trace $a + d - (d_1 + d_2)\sigma$ and determinant $h(\sigma) := d_1 d_2 \sigma^2 - (d_2 a + d_1 d)\sigma +$
 611 $(ad - bc)$. The first two conditions give stability at $\sigma = 0$; the trace is then negative for all $\sigma > 0$, so
 612 instability requires $h(\sigma) < 0$ for some $\sigma > 0$. The minimum of the convex parabola h is attained at
 613 $\sigma^* = (d_2 a + d_1 d)/(2d_1 d_2)$ and is negative precisely under the third condition. $\square$

614 Condition (12) requires the two channels to have *unequal* diffusivities of opposite-signed feedback—
 615 the classical short-range-activation / long-range-inhibition requirement. The operators fitted in Sec-
 616 tion 5 satisfy the first two conditions but not the third: the dispersion curve is monotonically decreas-
 617 ing (Fig. 7c), and $\mu^*(k) < 0$ for all $k > 0$ in every run.

618 *Remark 16* (Why the negative result is expected). The training objective (3) rewards making the
 619 *observed* configuration stationary. A Turing-unstable operator does the opposite: it renders the
 620 homogeneous state unstable so that structure grows spontaneously. Since the observed field is treated
 621 as a fixed point rather than as the outcome of an instability, the fitted operator is driven toward
 622 dissipativity. Detecting pattern-forming instability would require observations away from steady
 623 state, consistent with Corollary 12.

624 G Discretization invariance of the encoder

625 The encoder must map measurements on an arbitrary point set to a field. Let μ be the sampling
 626 measure on Ω and consider the kernel-integral operator

$$(\mathcal{K}v)(x) = \int_{\Omega} \kappa_{\theta}(y - x) \odot v(y) d\mu(y), \quad (13)$$

627 of which the graph layer is the empirical counterpart $(\mathcal{K}_N v)(x_i) = |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_{\theta}(x_j -$
 628 $x_i) \odot v(x_j)$.

629 **Proposition 17** (Monte-Carlo consistency). *If $\{x_j\}_{j=1}^N$ are i.i.d. draws from μ and $\kappa_{\theta} \odot v \in L^2(\mu)$,
 630 then for each x , $\mathbb{E}[(\mathcal{K}_N v)(x)] = (\mathcal{K}v)(x)$ and $\text{Var}[(\mathcal{K}_N v)(x)] = O(N^{-1})$; consequently $\|\mathcal{K}_N v -$
 631 $\mathcal{K}v\|_{L^2(\mu)} = O_P(N^{-1/2})$.*

632 *Proof.* The empirical operator is a sample mean of i.i.d. terms with finite second moment; unbi-
 633 asedness and the $O(N^{-1})$ variance are immediate, and the norm bound follows by Chebyshev and
 634 Fubini. $\square$

635 Proposition 17 is the sense in which the encoder is discretization-invariant: the population operator
 636 (13) depends on the geometry through $\kappa_{\theta}(y - x)$ alone and not on the sampling pattern, and the
 637 empirical operator converges to it as the sampling density increases, whatever the lattice. **This**
 638 **is an asymptotic statement about the encoder, and it is worth emphasising that it does not**
 639 **imply transfer of the fitted operator.** The transfer experiments of Section 5 show empirically that
 640 it does not: $\mathbf{D}_{\theta}$ and f_{θ} are fitted to one tissue’s stationary field, and Theorem 11 indicates they
 641 are unconstrained away from that field’s range. Discretization invariance of the *architecture* and
 642 portability of the *fit* are distinct properties, and only the former is established here.

643 **Proposition 18** (Splating consistency). *Let $w_{\sigma}(r) = e^{-r^2/2\sigma^2}$ and define the Nadaraya–Watson
 644 field $F_{\sigma}(u) = \sum_i w_{\sigma}(\|u - x_i\|)v_i / \sum_i w_{\sigma}(\|u - x_i\|)$. If $v_i = v(x_i)$ for $v \in C^2$ and the sampling
 645 density p is bounded away from zero, then $F_{\sigma}(u) = v(u) + O(\sigma^2) + O_P((N\sigma^2)^{-1/2})$.*

646 *Proof sketch.* Standard kernel-regression bias–variance decomposition in two dimensions: the bias
647 term is $\frac{\sigma^2}{2}(\Delta v + 2\nabla v \cdot \nabla \log p) + o(\sigma^2)$ and the variance is $O((N\sigma^2)^{-1})$. $\square$

648 The learnable bandwidth σ therefore trades a smoothing bias against sampling noise. Since the
649 bias term is proportional to Δv , splatting contributes a further diffusive smoothing on top of the
650 operator—a mechanism that plausibly compounds the over-smoothing reported in Section 5, and
651 one that the present work does not disentangle.

652 Complexity

653 Let N be the number of measured locations, G genes, C latent channels, $H \times W$ the lattice,
654 E the number of graph edges and T the integration horizon. The costs per forward pass are:
655 gene projection $O(NGd)$; graph layers $O(LEd)$; splatting $O(Nr^2C)$ for radius r ; spectral blocks
656 $O(BCHW \log HW)$; operator $O(TCHW \log HW)$; decoding $O(NCd + NGd)$. The operator
657 term is therefore linear in the horizon and quasi-linear in the lattice, and for the configurations used
658 here ($C=32$, $H=W=64$, $T=8$) it is under 4% of the forward cost—consistent with the parameter
659 count of Section 6, where the operator holds 0.43% of the weights yet accounts for the entire margin
660 over the strongest baseline.

661 H Empirical details and full result tables

662 This appendix collects the complete result tables underlying Section 5. Tables are placed inline with
663 [h] so that each appears with its heading.

664 H.1 Single-section model comparison

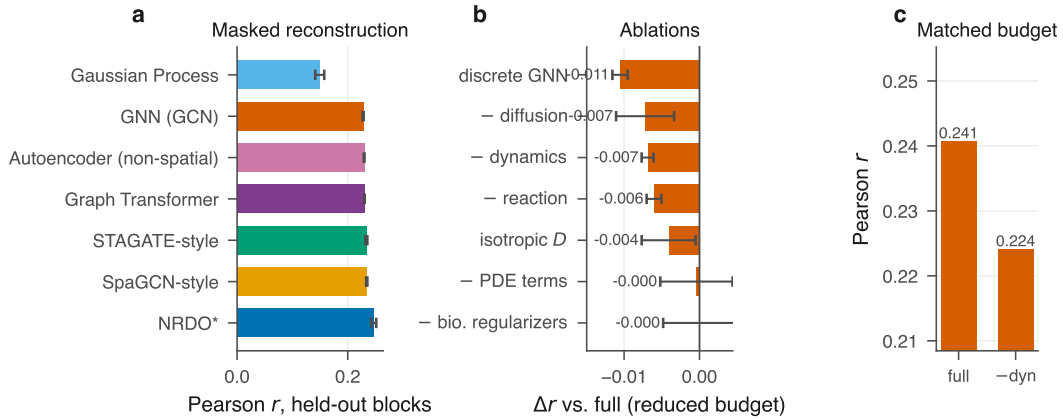


Figure 3: **Locating the effect** on the primary section, where all nine models are run. (a) Full model comparison. (b) Ablation deltas at the reduced budget. (c) The two decisive controls at the benchmark budget: disabling the operator ($T=0$) costs more than the full margin over the best baseline.

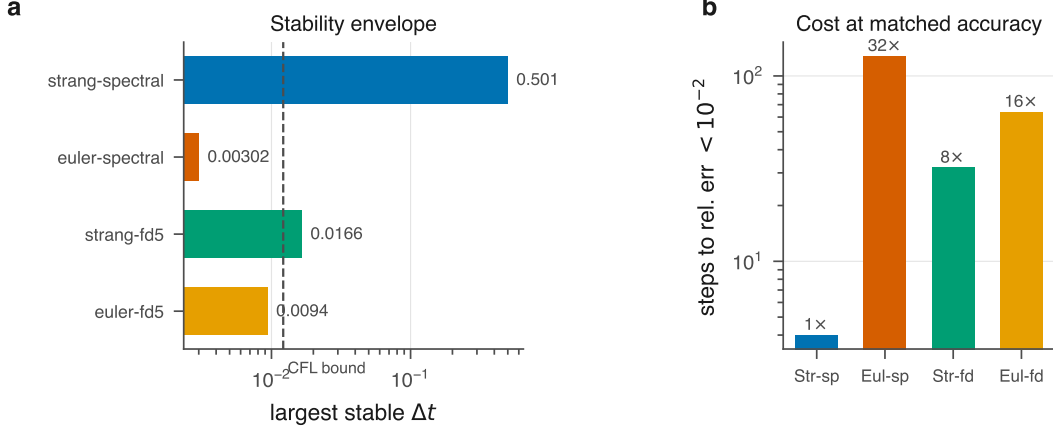


Figure 4: **What the exact spectral solve buys.** (a) Largest stable step size per scheme against the CFL bound for the explicit five-point Laplacian. (b) Steps required to integrate a fixed horizon to relative error $< 10^{-2}$, relative to the exponential scheme.

Table 2: Paired comparison of NRDO against each baseline, with the tissue section as the unit of analysis. Δ is the mean paired difference in Pearson r (positive favors NRDO) with a percentile bootstrap 95% CI over sections; d_z is Cohen’s effect size for paired designs; p is the Wilcoxon signed-rank test and p_{Holm} its Holm–Bonferroni correction across the baseline family.

baseline	n	Δr	95% CI	d_z	p	p_{Holm}	wins
SpaGE-style	17	+0.1416	[+0.1217, +0.1606]	3.36	1.5e-05	4.6e-05	17/17
Neural field (SIREN)	23	+0.0635	[+0.0514, +0.0765]	2.02	2.4e-07	1.7e-06	23/23
GP, multi-bandwidth	23	+0.0483	[+0.0402, +0.0569]	2.32	2.4e-07	1.7e-06	23/23
Tangram-style	23	+0.0194	[+0.0137, +0.0250]	1.37	4.5e-06	1.8e-05	21/23
GNN (GCN)	23	+0.0146	[+0.0097, +0.0199]	1.14	2.4e-06	1.2e-05	22/23
Autoencoder (non-spatial)	23	+0.0135	[+0.0077, +0.0197]	0.89	0.00021	0.00043	18/23
STAGATE-style	23	+0.0117	[+0.0058, +0.0180]	0.75	0.0027	0.0027	17/23

Table 3: Numerical behavior of the integrator variants. The largest stable step is measured by sweeping Δt over $[10^{-4}, 0.501]$ and testing boundedness over 300 steps; the CFL reference is $h^2/(4\lambda_{\max}) = 0.01215$ for the explicit five-point Laplacian. **strang-spectral never destabilized within the sweep** ($\Delta t \leq 0.501$), so its entry is a lower bound set by the grid rather than a measured threshold, consistent with the unconditional stability of Proposition 5. Cost is the number of steps required to integrate a fixed horizon to relative error $< 10^{-2}$, relative to the exponential scheme; the achieved error is reported alongside, since the step grid is coarse and the schemes do not land on the same accuracy. The empirical limit for euler-fd5 tracks the CFL bound, confirming the analysis.

scheme	largest stable Δt	vs. CFL	steps to 10^{-2}	achieved err.	cost
strang-spectral	≥ 0.501	$\geq 41\times$	4	8.4e-03	1x
euler-spectral	0.00302	0.25x	128	2.3e-03	32x
strang-fd5	0.0166	1.4x	32	9.1e-04	8x
euler-fd5	0.0094	0.77x	64	6.1e-03	16x

Table 4: Biological preservation of the reconstruction, averaged over 10 sections with reference annotations (Space Ranger clusters, Xenium clusters, and curated anatomical regions). Predicted expression at held-out locations is clustered and scored against the reference; ARI retention is the ratio to the score obtained from the measured data, so 1.0 would mean the reconstruction is as informative as the measurement. Marker AUROC asks whether the predicted field still discriminates a region using that region’s measured markers; k -NN preservation is the overlap of neighborhoods in measured versus predicted expression space; $|\Delta C|$ is the error in Geary’s C .

model	ARI $\uparrow$	ARI ret. $\uparrow$	NMI $\uparrow$	marker AUROC $\uparrow$	k -NN pres. $\uparrow$	$ \Delta C \downarrow$
GNN (GCN)	0.206 $\pm$ 0.147	-0.256 $\pm$ 2.333	0.349 $\pm$ 0.164	0.781 $\pm$ 0.055	0.157 $\pm$ 0.073	0.754 $\pm$ 0.061
STAGATE-style	0.205 $\pm$ 0.138	-0.135 $\pm$ 1.902	0.348 $\pm$ 0.157	0.787 $\pm$ 0.055	0.158 $\pm$ 0.074	0.739 $\pm$ 0.059
GP, multi-bandwidth	0.203 $\pm$ 0.135	-0.316 $\pm$ 2.243	0.361 $\pm$ 0.167	0.786 $\pm$ 0.091	0.187 $\pm$ 0.086	0.741 $\pm$ 0.071
Neural field (SIREN)	0.085 $\pm$ 0.055	0.151 $\pm$ 0.087	0.169 $\pm$ 0.104	0.698 $\pm$ 0.055	0.098 $\pm$ 0.053	0.534 $\pm$ 0.117
NRDO (ours)	0.224 $\pm$ 0.109	1.127 $\pm$ 2.040	0.373 $\pm$ 0.160	0.797 $\pm$ 0.059	0.188 $\pm$ 0.094	0.794 $\pm$ 0.064

Table 5: Processed spatial transcriptomics datasets. Locations and genes are counts after QC and gene selection; extent is the physical bounding box of the section.

dataset	technology	organism	locations	genes	extent (μm)	resolution
merfish_allen_01	MERFISH, 500-gene pane	Mus	23,376	500	4178 $\times$ 4097	single cell
merfish_allen_05	MERFISH, 500-gene pane	Mus	45,896	501	6127 $\times$ 5166	single cell
merfish_allen_10	MERFISH, 500-gene pane	Mus	42,070	503	8402 $\times$ 5441	single cell
merfish_allen_14	MERFISH, 500-gene pane	Mus	94,357	500	9048 $\times$ 5717	single cell
merfish_allen_18	MERFISH, 500-gene pane	Mus	44,309	500	7980 $\times$ 4825	single cell
merfish_allen_26	MERFISH, 500-gene pane	Mus	79,282	501	9460 $\times$ 5685	single cell
merfish_allen_30	MERFISH, 500-gene pane	Mus	91,017	501	9870 $\times$ 5843	single cell
merfish_allen_31	MERFISH, 500-gene pane	Mus	91,951	502	9837 $\times$ 6094	single cell
merfish_allen_36	MERFISH, 500-gene pane	Mus	115,675	503	10074 $\times$ 6559	single cell
merfish_allen_40	MERFISH, 500-gene pane	Mus	104,461	502	10000 $\times$ 6961	single cell
merfish_allen_42	MERFISH, 500-gene pane	Mus	75,433	503	9875 $\times$ 7086	single cell
merfish_allen_50	MERFISH, 500-gene pane	Mus	92,787	501	8806 $\times$ 6604	single cell
mosta_embryo_E10.5	Stereo-seq, DNA nanoba	Mus	18,360	2,099	3500 $\times$ 4950	bin50 (25 um)
mosta_embryo_E9.5	Stereo-seq, DNA nanoba	Mus	5,898	2,069	1975 $\times$ 2650	bin50 (25 um)
perturb_norman	Perturb-seq (CRISPRa),	Homo	111,659	2,073	–	single cell
visium_ffpe_human_prostate	Visium FFPE (55 um spo	Homo	4,369	2,088	6363 $\times$ 6706	spot (multi-cell)
visium_ffpe_mouse_brain	Visium FFPE (55 um spo	Mus	2,264	2,075	4959 $\times$ 5307	spot (multi-cell)
visium_human_breast	Visium (55 um spots, h	Homo	4,869	2,087	6395 $\times$ 6750	spot (multi-cell)
visium_human_heart	Visium (55 um spots, 1	Homo	331	2,051	5877 $\times$ 6040	spot (multi-cell)
visium_human_lymph_node	Visium (55 um spots, 1	Homo	4,025	2,073	5967 $\times$ 6744	spot (multi-cell)
visium_mouse_brain	Visium (55 um spots, 1	Mus	2,691	2,079	4822 $\times$ 6379	spot (multi-cell)
visium_mouse_brain_coronal	Visium (55 um spots, 1	Mus	2,896	2,082	4510 $\times$ 6689	spot (multi-cell)
visium_mouse_kidney	Visium (55 um spots, 1	Mus	1,433	2,079	4974 $\times$ 4894	spot (multi-cell)
xenium_mouse_brain	Xenium In Situ, 248-ge	Mus	36,362	248	5212 $\times$ 3128	single cell

Table 6: Masked spatial reconstruction across 23 tissue sections spanning four technologies; mean $\pm$ s.d. over sections, 2 seeds each. Excluded for incomplete coverage: Gaussian Process (1/23), SpaGE-style (17/23). Stars: Holm-corrected Wilcoxon signed-rank p against NRDO, section as the unit of analysis (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Win record and effect sizes: Table 2.

model	n	Pearson $r \uparrow$	SSIM $\uparrow$	$ \Delta I \downarrow$
STAGATE-style	23	0.165 $\pm$ 0.067**	0.319 $\pm$ 0.064*	0.764 $\pm$ 0.049***
Autoencoder (non-spatial)	23	0.163 $\pm$ 0.067***	0.256 $\pm$ 0.083***	0.755 $\pm$ 0.054***
GNN (GCN)	23	0.162 $\pm$ 0.065***	0.317 $\pm$ 0.063*	0.791 $\pm$ 0.047***
Tangram-style	23	0.157 $\pm$ 0.068***	0.338 $\pm$ 0.076	0.794 $\pm$ 0.073***
GP, multi-bandwidth	23	0.129 $\pm$ 0.056***	0.295 $\pm$ 0.055***	0.759 $\pm$ 0.064***
Neural field (SIREN)	23	0.113 $\pm$ 0.054***	0.251 $\pm$ 0.056***	0.593 $\pm$ 0.111***
NRDO (ours)	23	0.177 $\pm$ 0.068	0.329 $\pm$ 0.068	0.844 $\pm$ 0.048

Table 7: Held-out Pearson r under input corruption. Each model is re-trained under each corruption with identical splits, since the realistic deployment is training on the data one has. Columns are increasing corruption level (for *density*, decreasing fraction of the section retained).

model	L1	L2	L3	L4
<i>noise</i>				
STAGATE-style	0.208	0.195	0.177	0.124
NRDO (ours)	0.201	0.187	0.170	0.118
GNN (GCN)	0.198	0.185	0.168	0.115
<i>dropout</i>				
STAGATE-style	0.208	0.207	0.206	0.206
NRDO (ours)	0.201	0.196	0.197	0.193
GNN (GCN)	0.198	0.191	0.194	0.188
<i>density</i>				
NRDO (ours)	0.195	0.158	0.197	0.201
STAGATE-style	0.173	0.163	0.187	0.208
GNN (GCN)	0.169	0.149	0.169	0.198
<i>knn</i>				
STAGATE-style	0.208	0.208	0.209	0.209
NRDO (ours)	0.201	0.201	0.201	0.201
GNN (GCN)	0.199	0.198	0.197	0.195

665 H.2 Numerical behavior

666 H.3 Paired statistics, numerics and biology

667 H.4 Dataset inventory

668 H.5 Multi-section benchmark

669 H.6 Robustness

670 H.7 Spectral matching

671 Full discussion

672 Section 5 reports that our predicted fields are too smooth, and Section H.8 shows this is structural
673 rather than an artifact of the training budget. The natural remedy is a term matching the radially
674 averaged power spectrum of the predicted field to that of the measured field. We implement it,
675 sweep its weight, and report the trade-off (Table 8, Fig. 5; 2 seeds per point).

676 The form of the term matters more than its weight. Matching the *absolute* spectrum over all bands
677 leaves an amplitude degree of freedom the model can exploit without redistributing energy across
678 scales — a controlled check finds it penalizes a purely rescaled field more heavily than a genuinely
679 smoothed one — and its best setting costs -78% of the Pearson r to gain -9% of $|\Delta I|$. Matching
680 the spectrum *normalized* by total power over the upper half of the band removes that freedom and
681 penalizes the frequencies a dissipative operator actually loses. At $\lambda = 0.01$ it reaches $|\Delta I| = 0.710$,
682 better than the absolute form achieves at any weight, for -15% of the accuracy; the curve turns over
683 above that, so it is an optimum rather than a point on a slope.

684 The conclusion is a bound, not a fix. Across every weight and both forms the predicted autocorre-
685 lation bottoms out at $I_{\text{pred}} = 0.869$ against a measured 0.174 : weighting the term heavily enough
686 to destroy predictive accuracy still does not bring the reconstruction near the roughness of the data.
687 We keep $\lambda = 0$ elsewhere and conclude that the smoothness belongs to the operator class rather than
688 to the objective.

Table 8: Spectral matching: the trade-off between reconstruction accuracy and spatial fidelity as the term’s weight λ is swept, 2 seeds per point, on the primary section. $\lambda = 0$ is the model used everywhere else in this paper. The measured field has $I = 0.174$, so a faithful reconstruction would drive I_{pred} down toward that value. Two forms are compared: matching the absolute power spectrum across all radial bands, and matching the spectrum normalized by total power over the upper half of the band. The first leaves an amplitude degree of freedom that the model can exploit without redistributing energy across scales; the second removes it and penalizes only the frequencies a dissipative operator actually loses.

weight	Pearson $r \uparrow$	SSIM $\uparrow$	$ \Delta I \downarrow$	I_{pred}	$ \Delta C \downarrow$
<i>absolute spectrum, all bands</i>					
$\lambda = 0$	0.249 ± 0.005	0.405 ± 0.007	0.760 ± 0.004	0.934 ± 0.004	0.727 ± 0.003
$\lambda = 0.001$	0.240 ± 0.004	0.399 ± 0.004	0.758 ± 0.003	0.932 ± 0.003	0.728 ± 0.002
$\lambda = 0.01$	0.184 ± 0.007	0.368 ± 0.017	0.781 ± 0.018	0.955 ± 0.018	0.745 ± 0.010
$\lambda = 0.1$	0.123 ± 0.004	0.405 ± 0.002	0.720 ± 0.001	0.894 ± 0.000	0.711 ± 0.000
$\lambda = 1$	0.055 ± 0.007	0.401 ± 0.023	0.695 ± 0.003	0.869 ± 0.003	0.698 ± 0.003
<i>normalized spectrum, high band</i>					
$\lambda = 0.01$	0.210 ± 0.004	0.384 ± 0.001	0.710 ± 0.003	0.884 ± 0.003	0.702 ± 0.001
$\lambda = 0.1$	0.169 ± 0.000	0.274 ± 0.000	0.717 ± 0.002	0.892 ± 0.002	0.707 ± 0.000
$\lambda = 0.3$	0.135 ± 0.003	0.266 ± 0.001	0.712 ± 0.000	0.886 ± 0.000	0.703 ± 0.000
$\lambda = 1$	0.102 ± 0.000	0.256 ± 0.002	0.721 ± 0.002	0.896 ± 0.002	0.709 ± 0.002

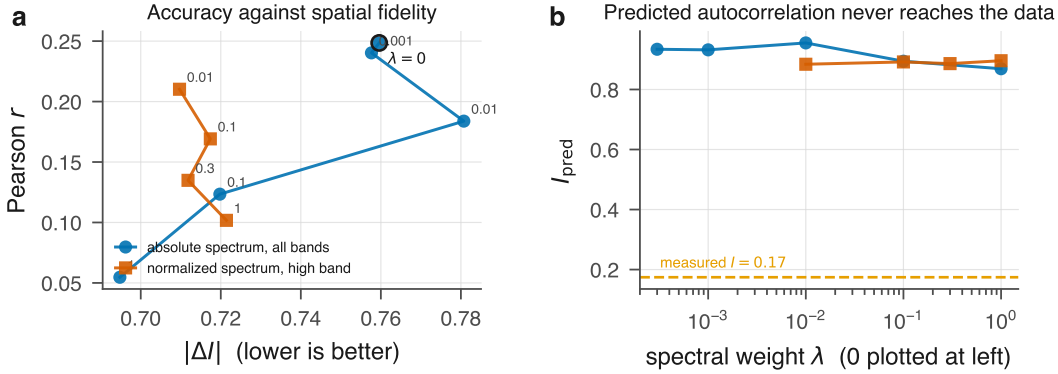


Figure 5: **Spectral matching.** (a) The trade-off: up and to the left is better. Points are labeled by weight λ ; $\lambda = 0$ is the model used everywhere else. (b) Predicted global autocorrelation against λ . No weight of either form brings I_{pred} near the measured value.

689 H.8 Converged single-section comparison

690 Full discussion

691 The benchmark trains every model for a fixed, reduced number of epochs, so its ordering could be
692 a fact about the budget rather than the models. We settle this on visium_mouse_brain, the section
693 where the ordering changes with budget, training every model to a shared early-stopping criterion
694 on the full section with 3 seeds (Appendix H, Table 9). NRDO attains 0.245 ± 0.006 against $0.232 \pm$
695 0.001 for the strongest baseline (STAGATE-style), winning on 3/3 seeds against every comparator
696 with $d_z = 1.99\text{--}2.84$, having *lost* to that baseline at the benchmark budget. The budget is therefore
697 conservative for NRDO here rather than neutral: Table 6 establishes consistency across anatomy at
698 a fixed budget, not that its margins are the converged margins.

699 Two findings survive convergence and matter more than the ordering. The non-spatial autoencoder
700 still matches the graph models (the best leads it by 0.004 Pearson r), so their near-tie is a property
701 of this task rather than of undertraining. And NRDO retains the largest error in Moran’s I (0.760
702 against 0.705), so the over-smoothing is structural. NRDO costs 653 s against 100–181 s per run on
703 one CPU.

Table 9: Converged comparison on visium_mouse_brain: the full section, no subsampling, trained to a shared early-stopping criterion on held-out validation rather than to a fixed epoch count, 3 seeds. d_z and the win record are paired over seeds against NRDO. This addresses whether the reduced-budget ordering of Table 6 is the converged ordering. It is not: at 200 epochs on this section NRDO scores below the STAGATE-style baseline, and at convergence it scores above it, so the benchmark budget is conservative for NRDO here rather than neutral. Two further points survive convergence — the non-spatial autoencoder still matches the graph models, so their near-tie is a property of the task and not of undertraining; and NRDO retains the largest error in Moran’s I , so the over-smoothing is structural. Wall time is single-CPU and is reported because NRDO is several times more expensive than every baseline.

model	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$	$ \Delta I \downarrow$	$ \Delta C \downarrow$	wall	d_z	wins
STAGATE-style	0.232 ± 0.001	0.255 ± 0.001	0.393 ± 0.002	0.708 ± 0.002	0.683 ± 0.001	181 s	1.99	3/3
Autoencoder (non-spatial)	0.228 ± 0.002	0.257 ± 0.001	0.392 ± 0.003	0.705 ± 0.004	0.680 ± 0.003	100 s	2.25	3/3
GNN (GCN)	0.227 ± 0.001	0.255 ± 0.001	0.391 ± 0.003	0.724 ± 0.008	0.692 ± 0.006	170 s	2.84	3/3
NRDO (ours)	0.245 ± 0.006	0.256 ± 0.007	0.399 ± 0.009	0.760 ± 0.009	0.726 ± 0.007	653 s	—	—

704 H.9 Null controls for the recovered diffusion length

Table 10: Null controls for the recovered diffusion length. Every row refits the identical architecture at the identical budget on the primary section. Shuffling coordinates destroys all spatial structure while preserving every gene’s marginal; those rows reach $r \approx 0$ yet recover the length scale of the *untrained* model, so the quantity is close to its initialization rather than inferred from tissue. The bandwidth and lattice sweeps show why neither is the mechanism: the length is invariant in microns across a fourfold change in pitch while the same quantity in lattice cells moves by the same factor.

condition	length (μm)	length (cells)	held-out r
measured tissue, $H=64$	374.6 ± 1.7	3.76	0.248
splat $\sigma = 0.5$ cells	375.5 ± 0.9	3.77	0.247
splat $\sigma = 2$ cells	377.7 ± 1.4	3.79	0.247
splat $\sigma = 4$ cells	376.4 ± 0.7	3.77	0.247
lattice $H=32$ (pitch $199 \mu\text{m}$)	376.3 ± 0.2	1.89	0.245
lattice $H=128$ (pitch $50 \mu\text{m}$)	378.2 ± 5.6	7.58	0.243
coordinates shuffled, $H=64$	404.5 ± 0.1	4.06	0.004
coordinates shuffled, $H=128$	404.7 ± 0.1	8.12	0.004

705 The rows above the rule refit the model on measured tissue under varied encoder and discretization
706 settings; the rows below refit it on sections whose expression has been permuted across positions.
707 The shuffled fits attain $r \approx 0$, as they must, and yet return a length scale within -0.06% of the
708 untrained model’s $405 \mu\text{m}$. Measured tissue moves the same quantity by -7.4% . Section 5 reports
709 what this does and does not license.

710 H.10 Ablation sweep

711 Full sweep at the reduced budget; the two matched-budget controls are reported in Section 6.

712 H.11 Cross-tissue and cross-resolution transfer

713 H.12 Developmental forecasting

714 Horizon sweep for $E9.5 \rightarrow E10.5$. Consecutive MOSTA stages are distinct embryos, so the compar-
715 ison is at the level of the rasterized field after isotropic normalization to a common frame.

Table 11: Ablations on the mouse-brain Visium section. Each variant removes exactly one ingredient with all else held fixed. Δr is the change in held-out Pearson r relative to the full model. The *– dynamics* row is the critical control: it disables the operator entirely ($T=0$) while leaving its parameters allocated, so the two rows are exactly parameter-matched. **This table uses a smaller training budget than the main benchmark** (fewer epochs and seeds); it is internally consistent, but absolute values are not comparable across the two tables.

variant	params	Pearson $r \uparrow$	Δr
Full model	2.20M	0.233 $\pm$ 0.005	–
– dynamics ($T=0$)	2.20M	0.227 $\pm$ 0.001	-0.007
– diffusion term	2.20M	0.226 $\pm$ 0.004	-0.007
– reaction term	2.19M	0.227 $\pm$ 0.001	-0.006
– PDE constraints	2.20M	0.233 $\pm$ 0.005	-0.000
– biological regularizers	2.20M	0.233 $\pm$ 0.005	-0.000
isotropic D	2.20M	0.229 $\pm$ 0.004	-0.004
state-dependent D	2.20M	0.236 $\pm$ 0.004	+0.003
discrete GNN operator	0.61M	0.223 $\pm$ 0.001	-0.011
latent $C=8$	1.07M	0.213 $\pm$ 0.005	-0.021
latent $C=16$	1.30M	0.220 $\pm$ 0.002	-0.013
latent $C=64$	5.77M	0.237 $\pm$ 0.000	+0.004

Table 12: Cross-tissue, cross-species transfer: adult mouse brain $\rightarrow$ human breast carcinoma. Shared gene vocabulary: 390 genes. Mean $\pm$ s.d. over seeds.

model	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$
<i>source (in-domain)</i>			
NRDO (ours)	0.214 $\pm$ 0.004	0.252 $\pm$ 0.004	0.369 $\pm$ 0.006
STAGATE-style	0.214 $\pm$ 0.000	0.249 $\pm$ 0.000	0.362 $\pm$ 0.001
GNN (GCN)	0.206 $\pm$ 0.001	0.250 $\pm$ 0.000	0.363 $\pm$ 0.004
Gaussian Process	0.146 $\pm$ 0.002	0.291 $\pm$ 0.004	0.289 $\pm$ 0.003
<i>target (zero-shot)</i>			
STAGATE-style	0.071 $\pm$ 0.006	0.403 $\pm$ 0.001	0.285 $\pm$ 0.008
Gaussian Process	0.070 $\pm$ 0.004	0.453 $\pm$ 0.001	0.311 $\pm$ 0.008
GNN (GCN)	0.056 $\pm$ 0.009	0.415 $\pm$ 0.006	0.250 $\pm$ 0.004
NRDO (ours)	-0.005 $\pm$ 0.010	0.504 $\pm$ 0.016	0.165 $\pm$ 0.009
<i>target (decoder-only fine-tune)</i>			
STAGATE-style	0.129 $\pm$ 0.009	0.386 $\pm$ 0.001	0.387 $\pm$ 0.005
GNN (GCN)	0.112 $\pm$ 0.005	0.390 $\pm$ 0.003	0.378 $\pm$ 0.006
NRDO (ours)	0.067 $\pm$ 0.021	0.398 $\pm$ 0.005	0.300 $\pm$ 0.023
Gaussian Process	–	–	–
<i>target (in-domain oracle)</i>			
STAGATE-style	0.149 $\pm$ 0.000	0.375 $\pm$ 0.000	0.433 $\pm$ 0.003
GNN (GCN)	0.145 $\pm$ 0.001	0.377 $\pm$ 0.000	0.426 $\pm$ 0.001
NRDO (ours)	0.136 $\pm$ 0.002	0.397 $\pm$ 0.002	0.329 $\pm$ 0.003
Gaussian Process	0.068 $\pm$ 0.002	0.459 $\pm$ 0.001	0.301 $\pm$ 0.008
<i>target (training-mean floor)</i>			
mean	-0.000	0.408	0.342

Table 13: Cross-resolution transfer: 55 μm Visium spots $\rightarrow$ single-cell Xenium. Shared gene vocabulary: 248 genes. Mean $\pm$ s.d. over seeds.

model	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$
<i>source (in-domain)</i>			
NRDO (ours)	0.261 ± 0.006	0.246 ± 0.001	0.392 ± 0.005
STAGATE-style	0.246 ± 0.000	0.247 ± 0.001	0.370 ± 0.002
GNN (GCN)	0.236 ± 0.002	0.249 ± 0.001	0.363 ± 0.002
Gaussian Process	0.158 ± 0.005	0.291 ± 0.004	0.278 ± 0.011
<i>target (zero-shot)</i>			
STAGATE-style	0.242 ± 0.002	1.780 ± 0.002	0.631 ± 0.003
GNN (GCN)	0.234 ± 0.002	1.798 ± 0.005	0.618 ± 0.002
Gaussian Process	0.183 ± 0.008	2.114 ± 0.032	0.577 ± 0.003
NRDO (ours)	0.124 ± 0.017	1.855 ± 0.030	0.544 ± 0.016
<i>target (in-domain oracle)</i>			
NRDO (ours)	0.277 ± 0.005	1.745 ± 0.006	0.667 ± 0.004
STAGATE-style	0.269 ± 0.001	1.756 ± 0.002	0.677 ± 0.002
GNN (GCN)	0.267 ± 0.001	1.760 ± 0.001	0.675 ± 0.001
Gaussian Process	0.165 ± 0.003	2.142 ± 0.022	0.559 ± 0.017
<i>target (training-mean floor)</i>			
mean	-0.000	1.877	0.398

Table 14: Developmental forecasting on Stereo-seq: the E9.5 field is encoded, integrated forward for T operator steps, and scored against the measured E10.5 field. **Consecutive stages are different embryos**, so there is no cell-to-cell correspondence and the comparison is made at the level of the rasterized field after isotropic normalization to a common frame; it inherits the error of that coarse registration. *Persistence* predicts $E10.5 = E9.5$ and is the reference the operator must beat for its dynamics to carry temporal information.

model	horizon	field $r \uparrow$	field RMSE $\downarrow$	genes with $r > 0$
Persistence (no dynamics)	–	0.064	0.346	61%
Spatial-mean predictor	–	0.000	0.257	4%
NRDO	$T=0$	0.082 ± 0.001	0.320	74%
NRDO	$T=2$	0.089 ± 0.000	0.305	72%
NRDO	$T=4$	0.092 ± 0.000	0.297	73%
NRDO	$T=8$	0.094 ± 0.000	0.289	72%
NRDO	$T=16$	0.095 ± 0.001	0.283	72%
NRDO	$T=32$	0.090 ± 0.003	0.285	70%

716 H.13 In-silico morphogen source

717 Split-half protocol of Section 5.6: the perturbation direction is defined from a random half of each
718 pathway and enrichment scored on the held-out half, against a permutation null over matched gene sets.

Table 15: In-silico morphogen source (*bead implant*). The latent perturbation direction is defined from a random half of each pathway’s genes; the table reports where the *held-out* half ranks in the response magnitude distribution (lower is stronger; 50% is chance). p is a permutation test over random gene sets of matched size. Response half-width is the distance at which the predicted response falls to half its peak. $*p < 0.05$.

pathway	genes	held-out rank (%)	null (%)	p	half-width (μm)
BMP	22	48.6 ± 6.0	49.7	0.475	575 ± 154
FGF	19	72.6 ± 0.8	50.3	0.992	618 ± 58
SHH	15	56.6 ± 6.7	50.4	0.698	466
WNT	26	56.4 ± 3.9	49.9	0.762	649 ± 143

719

720 **H.14 Perturb-seq consistency**

721 An out-of-context probe: Norman et al. profiled non-spatial K562 cells whereas the operator is fitted
 722 to tissue. With $n = 7$ testable perturbations the comparison is underpowered and we report it as
 inconclusive.

Table 16: Agreement between the model’s counterfactual gene–gene responses and measured CRISPRa responses from Norman et al. (2019). **This is an out-of-context probe:** the Perturb-seq data are non-spatial K562 cells while the operator is fitted to human breast tissue, so it tests whether the reaction module encodes generic transcriptional coupling, not whether it recovered tissue-specific signaling. Null is a within-perturbation label permutation; p is a Wilcoxon signed-rank test against that null.

model	#perturb.	Spearman ρ	null	% positive	p
STAGATE-style	7	0.010 ± 0.000	+0.004	64%	0.633
NRDO (ours)	7	-0.018 ± 0.009	-0.009	57%	0.844

723

724 **H.15 Latent fields**

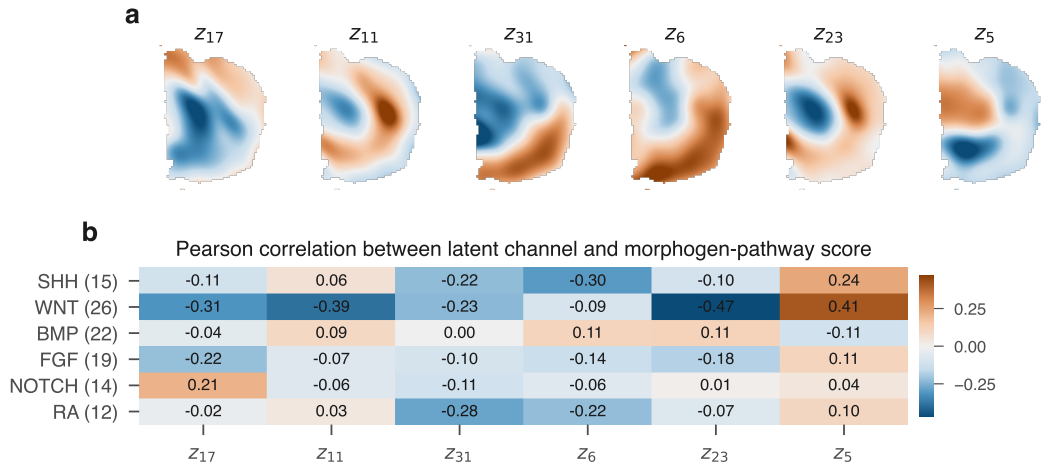


Figure 6: Latent channels of the relaxed field (a) and their correlation with canonical morphogen-pathway scores (b). Section 5.6 shows these correlations do not survive a causal split-half test.

725 **I Reproducibility**

726 All datasets download automatically with a recorded SHA256 manifest; every table and figure in this
 727 paper is regenerated from run artifacts by a single command; configurations, seeds and checkpoints
 728 are released.

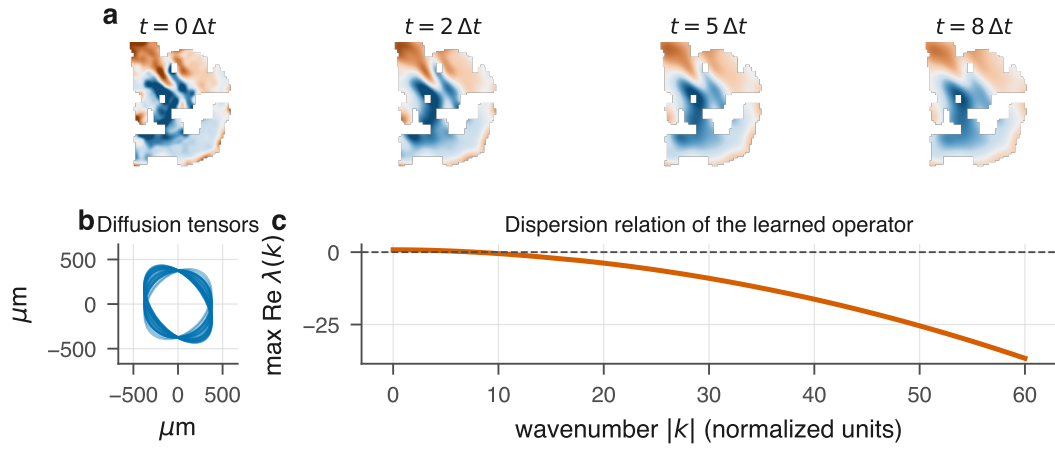


Figure 7: **The fitted operator.** (a) A latent channel relaxing under the learned dynamics. (b) Per-channel diffusion tensors as length-scale ellipses, showing the anisotropy is used. (c) Dispersion relation of the linearized operator, which decreases monotonically in $|k|$ and so supports no Turing band.